

Tatvik Sinha

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Education

B.Tech in Computer Science & Engineering (AI & ML)

CGPA: 9.12/10

Vellore Institute of Technology, Bhopal Campus

2024 – 2028

Class XII (CBSE) – 88%

Swami Vivekananda School, 2022

Class X (ICSE) – 92.3%

St. George's College, 2020

Technical & Consulting Skills

Cloud & Consulting: Microsoft Azure, Technical Solution Design, Risk Mitigation, Agile/Scrum, Stakeholder Management

Programming Languages: Python, C++, Java, JavaScript, SQL

Frontend Development: React.js, HTML5, CSS3, Responsive UI, State Management

Backend Development: Node.js, Express.js, REST APIs, Authentication, JWT, Microservices

Artificial Intelligence: LLMs, RAG, Prompt Engineering, Semantic Search, NLP, Predictive Modeling

Databases & DevOps: PostgreSQL, SQL, Vector Databases, Docker, Git, GitHub Actions, Linux

Experience

Machine Learning Intern, FlyRank AI

July 2026 – Present

- Consulted with cross-functional stakeholders to define business requirements, architecting predictive ML models that aligned directly with strategic data-driven goals.
- Implemented scalable, cloud-ready AI solutions (NLP, automated analytics), identifying opportunities to accelerate technology adoption and mitigate deployment risks.
- Executed advanced data preprocessing and optimization, communicating technical insights to stakeholders to drive business satisfaction.

Backend Developer Intern, Shelter Zone NGO

May 2026 – July 2026

- Designed and implemented technical solutions via scalable RESTful APIs (Node.js) to address core operational constraints for NGO outreach initiatives.
- Cultivated stakeholder relationships to gather technical requirements, managing project dependencies to deliver seamless frontend-to-backend integrations.
- Designed optimized PostgreSQL schemas and implemented robust JWT authentication protocols to ensure strict data privacy and risk mitigation.

Projects

BiFin – Enterprise Banking Intelligence Platform

React, FastAPI, Python, Scikit-Learn, Optuna, Docker

- Designed a full-stack enterprise solution offering (FastAPI, React) aligned with business goals to predict customer churn and propose retention mitigations.
- Led project planning by defining technical constraints and engineering a robust data ingestion layer to handle missing transaction data and mitigate execution risks.
- Designed an automated ML pipeline with Optuna tuning and SHAP explainability to provide transparent pre-sales technical input to simulated business clients.
- Deployed the unified application using Docker Compose, creating seamless communication between the React frontend and Python microservices.
- Achieved **87.40% accuracy** and **0.8715 ROC-AUC** via leakage-free cross-validation, exceeding project success metrics.

Resa-AI – Scientific NLP & ML Extraction Pipeline

Django, Python, spaCy, XGBoost, Scikit-Learn, Docker

- Architected a scalable ML solution for autonomous PDF classification, successfully translating complex academic constraints into deployable intellectual property (IP).
- Engineered a robust preprocessing pipeline integrating TF-IDF vectorization and feature scaling to accelerate cloud technology adoption.
- Automated the downloading and processing of HuggingFace datasets, actively identifying gaps in existing methodologies to improve solution accuracy.
- Identified execution risks and dependencies, proposing mitigations via comprehensive Pytest suites and robust CI/CD integration.
- Evaluated multiple algorithms, achieving **83.27% accuracy** and **0.9278 ROC-AUC** on **2,400+ sentences**, deployed via Docker for high customer satisfaction.

Achievements

- Awarded the NPTEL Gold Medal in Cloud Computing by IIT Kharagpur, ranking among the top 2% of learners globally.
- Selected among the top 15,000 candidates nationwide in the prestigious Google BigCode competition.